

XGBOOST REGRESSOR

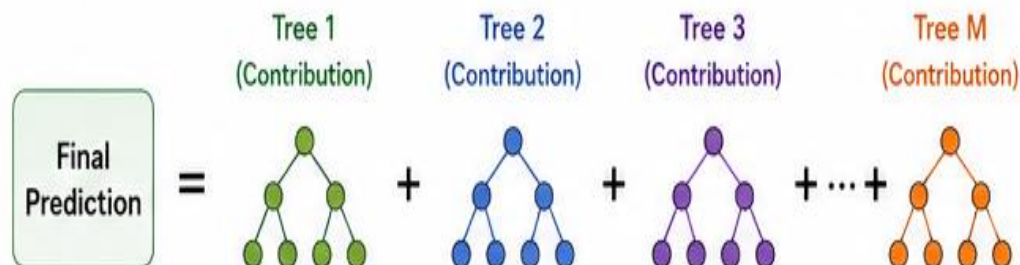
Gradient Boosting Regressor – How It Works

Gradient Boosting builds an additive model in a forward stage-wise fashion. In each stage, a regression tree is fit on the negative gradient of the given loss function.

Additive Model

1. What is an Additive Model?

The model prediction is the sum of the outputs of multiple trees.



$$\hat{y}(x) = f_0(x) + \nu \sum_{m=1}^M h_m(x)$$

$\hat{y}(x)$: Final prediction for input x

$f_0(x)$: Initial prediction (usually the mean of target)

$h_m(x)$: Prediction from m -th tree

ν : Learning rate (shrinkage factor, $0 < \nu \leq 1$)

M : Total number of trees

2. How Additive Model Works (Example)

Suppose the actual value for a data point is 10.

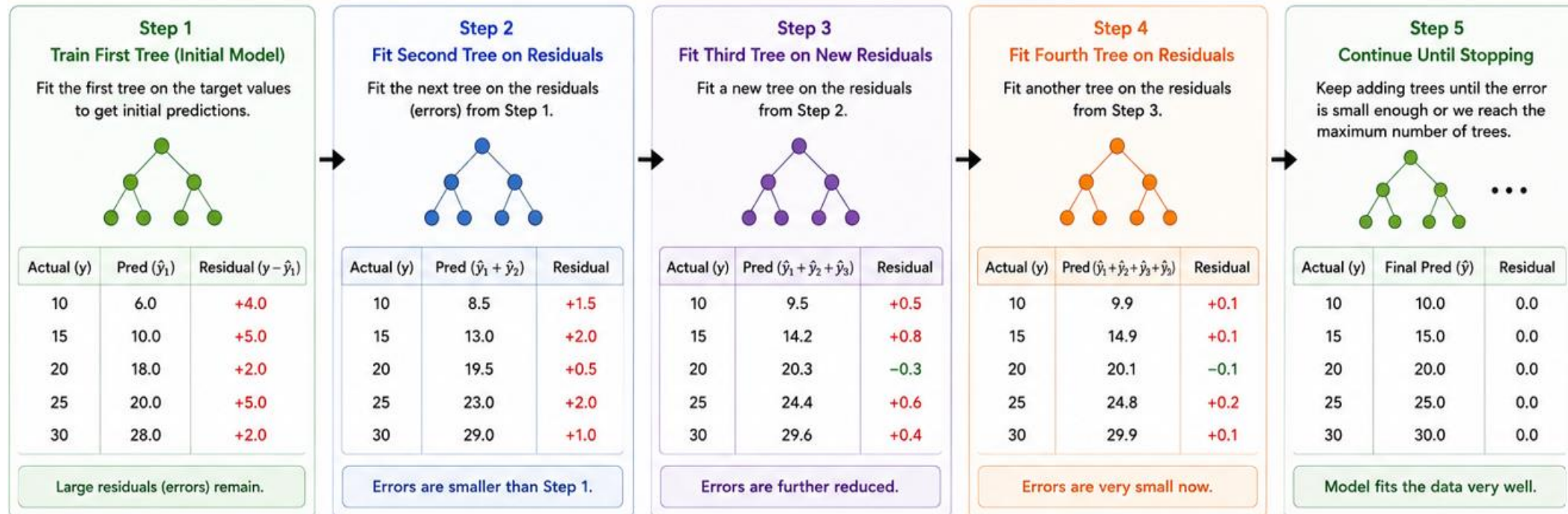
We start with an initial prediction and add trees one by one.

Step	Model	Tree Output (Contribution)	Total Prediction	Error (Actual - Prediction)
Initial	$f_0(x)$	5.0	5.0	+5.0
After Tree 1	$f_0(x) + \nu h_1(x)$	+2.0	7.0	+3.0
After Tree 2	$f_0(x) + \nu h_1(x) + \nu h_2(x)$	+1.5	8.5	+1.5
After Tree 3	$f_0(x) + \nu h_1(x) + \nu h_2(x) + \nu h_3(x)$	+1.0	9.5	+0.5
After Tree 4	$f_0(x) + \nu \sum_{m=1}^4 h_m(x)$	+0.5	10.0	0.0

Each tree adds a small correction (positive or negative) to reduce the error.
After many trees, the prediction becomes very close to the actual value.

Forward stage- wise fashion

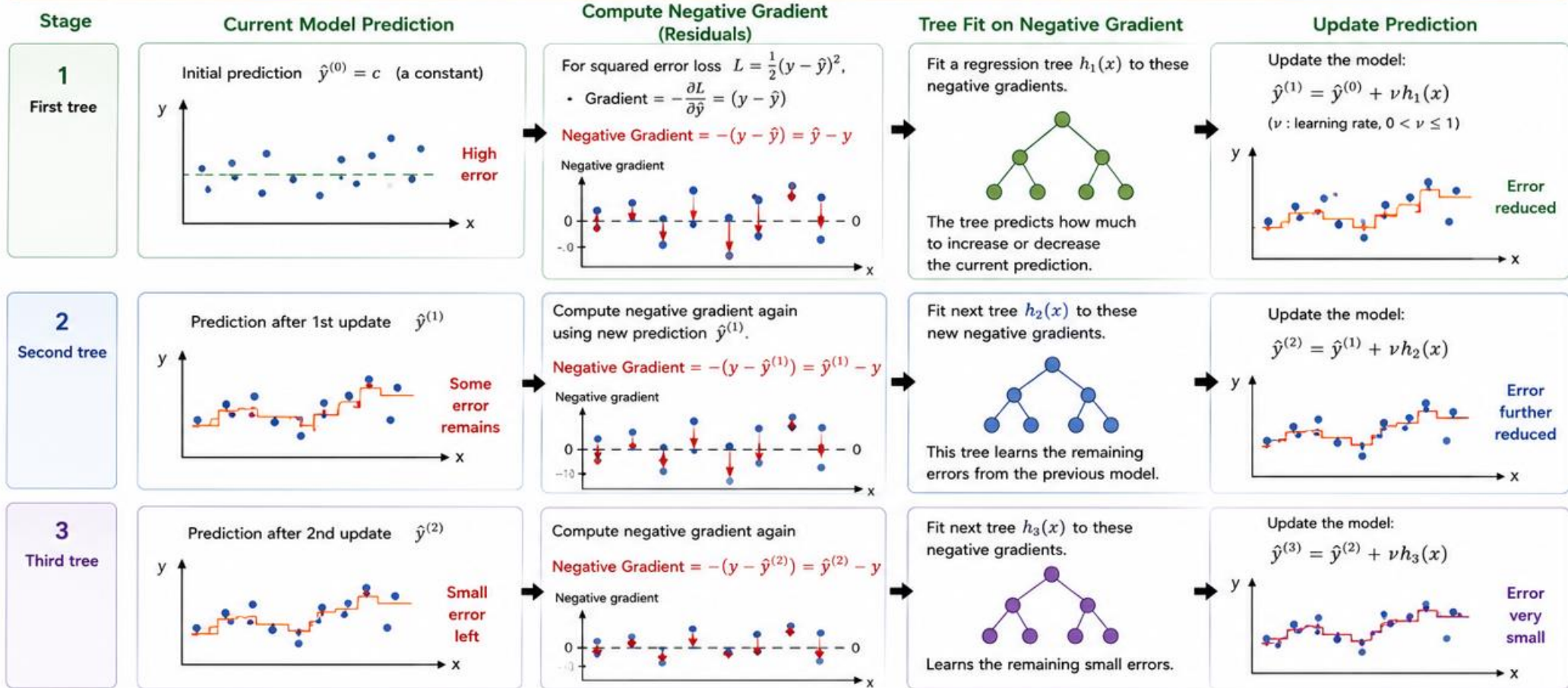
Forward stage-wise fashion means the model is built **one step at a time** (one tree at a time).
At each step, a new tree is added to **correct the errors** (residuals) made by all the previous trees.



In Each Stage, a Tree is Fit on the Negative Gradient

Gradient Boosting does not fit a tree on the target values (y).

Instead, at each stage, it fits a regression tree on the **negative gradient** of the loss function, which tells the **direction and amount** by which the current prediction should be corrected.



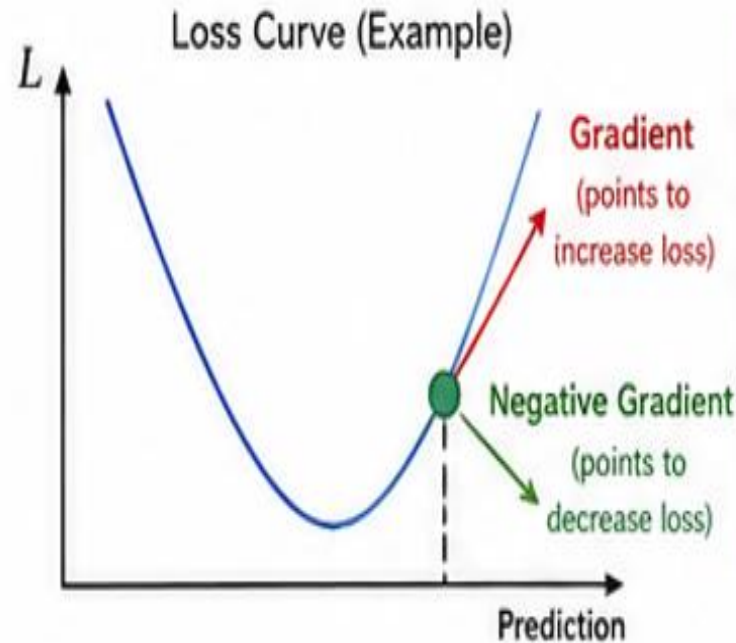
4. Optimization with Differentiable Loss Function

Loss Function (L)

Measures how wrong the predictions are.

Examples:

- Mean Squared Error
- Mean Absolute Error
- Huber Loss
- etc.



Why Negative Gradient?

Gradient shows the direction of steepest increase in loss.

We use the negative gradient because it shows the direction of steepest decrease in loss.

workflow

